TASK 3: DEEP-DIVE ANALYSIS & INTERACTIVE DASHBOARDING

ApexPlanet Software Pvt. Ltd. — Data Analytics Internship

1. Introduction

This project focuses on analyzing e-commerce sales data using Microsoft Power BI. The objective is to identify business trends, evaluate sales performance, and develop an interactive dashboard that supports data-driven decision-making.

2. Project Objectives

- Define and analyze key business performance indicators.
- Analyze sales performance across product categories.
- Identify high-performing cities.
- Understand monthly sales trends.
- Analyze customer sales contribution by gender.
- Build an interactive Power BI dashboard.

3. Dataset Description

The dataset contains 1,000 records and 12 columns related to e-commerce transactions.

Key attributes include:

- Order ID
- Customer ID and Customer Name
- Order Date
- Product and Category
- City
- Gender
- Quantity
- Unit Price
- Total Sales

The dataset was checked for missing values and prepared for analysis.

4. Tools and Technologies

- Microsoft Power BI
- DAX (Data Analysis Expressions)
- Data Cleaning
- Data Visualization
- Exploratory Data Analysis

5. Key Performance Indicators (KPIs)

KPI	Result
Total Sales	139.40M
Total Orders	992
Total Customers	947
Total Quantity	Approximately 5K
Average Order Value	140.52K

6. Deep-Dive Analysis

A. Category-wise Sales Analysis

Electronics generated the highest sales, approximately 51M, followed by Education at 25M.

This indicates that Electronics is the leading sales category in the dataset.

B. City-wise Sales Analysis

Patna recorded the highest sales at approximately 20.8M, followed by Kolkata at 18.9M.

This analysis helps identify cities contributing significantly to overall sales.

C. Monthly Sales Trend

Monthly sales remained approximately between 9M and 13M during most of 2025.

The trend helps monitor changes in sales performance over time and identify periods of higher or lower sales.

D. Gender-wise Sales Analysis

Male customers contributed approximately 51.98% of sales, while female customers contributed approximately 48.02%.

The sales contribution is relatively balanced across both customer groups.

7. Interactive Dashboard Features

The Power BI dashboard includes:

- KPI Cards
- Sales by Category
- Sales by City
- Monthly Sales Trend
- Gender-wise Sales Analysis
- Date Filter
- Category Filter

- City Filter

These interactive filters allow users to explore sales performance from different perspectives.

8. Key Business Insights

1. Electronics is the highest-performing product category.
2. Patna is the leading city in terms of sales.
3. Monthly sales show relatively consistent performance throughout most of 2025.
4. Sales contribution is nearly balanced between male and female customers.
5. Interactive filtering makes it easier to explore business performance across categories, cities, and dates.

9. Conclusion

This project demonstrates how Power BI can transform raw e-commerce data into meaningful business insights.

The interactive dashboard provides a clear overview of sales performance and supports business analysis through KPIs, visualizations, and filters.

10. Future Scope

- Profit and discount analysis
- Customer segmentation
- Customer retention analysis
- Sales forecasting
- Advanced business performance monitoring

Project Deliverables

- Power BI Interactive Dashboard (.pbix)
- Cleaned Dataset (.csv)
- Project Documentation (.pdf)
- GitHub Repository